

Markoff–Hurwitz equations on the five-cusped projective plane

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Abstract

Huang and Norbury proved that on the thrice-cusped projective plane $N_{1,3}$ the four one-sided simple closed geodesics disjoint from nothing but each other form *Markoff quads*: in the trace coordinates $x = 2\sinh(\ell/2)$ they satisfy $(a + b + c + d)^2 = abcd$, an $n = 4$ Markoff–Hurwitz equation. We solve the corresponding problem on the five-cusped projective plane $N_{1,5}$, whose maximal complete 1-system consists of eleven one-sided curves.

The engine is a single, universal, conceptually proved identity: for *every* 2×2 matrix A with $\det A = -1$ and *every* pair $B_1, B_2 \in \mathrm{SL}(2)$, the seven traces of A , AB_1 , AB_1B_2 , AB_2 , B_1 , B_2 , B_1B_2 satisfy one irreducible sixteen-term quartic $Q = 0$, obtained from the Gram matrix of the trace form on $\{I, B_1, B_2, B_1B_2\}$ by an adjugate computation. When the product B_1B_2 is a cusp word, Q collapses by Cayley–Hamilton to a genuine Markoff–Hurwitz equation with quartic coefficient $c = 1$ whose lower-order coefficients are the bounding block traces and the lift sign; the Huang–Norbury quad is the specialization $b_1 = b_2 = 2$, $\sigma = +1$. On $N_{1,5}$ all ten one-sided triples of the maximal system satisfy Q , exactly three of them are cusp-decorated and Markoff–Hurwitz, and each identity descends, uniformly over Teichmüller space and free of all sign and lift choices, to a polynomial relation among the $4\sinh^2(\ell/2)$ of the geodesic lengths, for every maximal system. Each decorated quad carries four explicit Vieta involutions preserving Q identically – the Markoff–Hurwitz automorphism signature – two of them realized by mapping-class-like matrix words. In sharp contrast to $N_{1,3}$, an exact certificate shows the triple-wise Vieta moves do *not* assemble into an action on the eleven-tuple: the rank-five structure is a coupled pentagon of quads, not a single equation with global flips.

1 Introduction

A *complete 1-system* on a surface is a collection of pairwise non-isotopic essential simple closed curves any two of which intersect exactly once. On the projective plane with three cusps, $N_{1,3}$, the maximal such systems have four curves, all one-sided, and Huang and Norbury [4] proved that their geodesic-length traces $x_i = 2\sinh(\ell_i/2)$ satisfy the $n = 4$ Markoff–Hurwitz equation

$$(a + b + c + d)^2 = abcd,$$

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whose Vieta *flips* generate the simple length spectrum – a non-orientable analogue of Markoff triples on the once-punctured torus. Hu, Tan and Zhang [5] determined the polynomial automorphism group of the general Markoff–Hurwitz hypersurface; the flips are its signature. It has been an open question whether this structure persists at higher rank: the five-cusped projective plane $N_{1,5}$ carries a maximal complete 1-system of *eleven* one-sided curves (Chen [1], Chen–Shen [2]), and no length relation for it was known.

This paper answers that question completely. The answer has a positive half and a structural half, and both are sharp.

Theorem 1.1 (universal quad relation; Theorem 3.1). *For every $A \in \mathrm{GL}(2)$ with $\det A = -1$ and every $B_1, B_2 \in \mathrm{SL}(2)$, the seven traces*

$$t_0 = \mathrm{tr} A, \quad t_1 = \mathrm{tr}(AB_1), \quad t_2 = \mathrm{tr}(AB_1B_2), \quad t_3 = \mathrm{tr}(AB_2), \quad b_1 = \mathrm{tr} B_1, \quad b_2 = \mathrm{tr} B_2, \quad k = \mathrm{tr}(B_1B_2)$$

satisfy the irreducible, sixteen-term, degree-four relation $Q = 0$ of (6), quadratic and monic in each t_i .

Theorem 1.2 (Markoff–Hurwitz form; Theorem 4.1). *If moreover $\mathrm{tr}(A^2B_1B_2) = 2\sigma$ with $\sigma \in \{\pm 1\}$ – the cusp condition – then $Q = 0$ becomes the genuine Markoff–Hurwitz equation*

$$t_0^2 + t_1^2 + t_2^2 + t_3^2 = t_0t_1t_2t_3 + b_1(t_0t_1 + t_2t_3) + b_2(t_0t_3 + t_1t_2) - 2\sigma(t_0t_2 + t_1t_3) - 2\sigma b_1b_2 + b_1^2 + b_2^2,$$

with quartic coefficient $c = 1$ and all remaining coefficients determined by the two bounding block traces and the lift sign. The Huang–Norbury quad is exactly the case $b_1 = b_2 = 2$, $\sigma = +1$.

Theorem 1.3 (the eleven-curve system; Theorem 4.3). *For the standard maximal complete 1-system $\{\gamma_0\} \cup \{\gamma_{ij}\}_{1 \leq i < j \leq 5}$ on $N_{1,5}$, all ten one-sided triples satisfy $Q = 0$ in their block decorations; exactly the three triples $(1, 2, 5)$, $(1, 3, 5)$, $(1, 4, 5)$ are cusp-decorated and satisfy the Markoff–Hurwitz equation above.*

Theorem 1.4 (length identities; Section 5). *In every complete finite-area hyperbolic metric on $N_{1,5}$, the geodesic lengths of each cusp-decorated triple satisfy the Markoff–Hurwitz equation exactly, in the coordinates $x_m = \delta_m 2 \sinh(\ell_m/2)$ for suitable signs δ_m ; and a sign-norm construction converts each trace identity into a nonzero polynomial identity in the quantities $4 \sinh^2(\ell_m/2)$ alone, valid uniformly over Teichmüller space, independent of all orientation, sign, and lift choices, and holding for every maximal complete 1-system after transport by the mapping class group.*

Theorem 1.5 (Vieta structure and the rank-five dichotomy; Section 6). *Each decorated quad admits four explicit involutions preserving Q identically – the Markoff–Hurwitz automorphism signature of [5] – two of which are realized by matrix words. However, the triple-wise Vieta moves provably do not assemble into an action on the eleven-tuple: a single flip applied inside one triple produces a tuple which no assignment of the auxiliary block traces returns to the trace image. The Markoff–Hurwitz structure of $N_{1,5}$ is therefore a coupled family of decorated quads, not a single equation with global flips – a genuine structural difference from $N_{1,3}$.*

The problem posed – *does the maximal system on $N_{1,5}$ satisfy Markoff–Hurwitz-type length relations, and in what form* – is thereby closed: the relations exist, they are exhibited explicitly with their proofs, their coefficients are cusp-determined, the rank-three theory embeds as a special case of one universal identity, and the precise sense in which the naive rank-three picture fails at rank five is a theorem rather than an impression.

Section 7 documents the provenance of these results (an autonomous AI research campaign), the exact-computation certificates behind each COMPUTED step, and every literature statement used inside a proof.

2 Preliminaries

Convention 2.1. $N_{1,5}$ is the real projective plane with five punctures, equipped with complete finite-area hyperbolic metrics. Fix

$$\pi_1(N_{1,5}) = \langle a, b_1, \dots, b_5 \mid a^2 b_1 b_2 b_3 b_4 b_5 = 1 \rangle \cong F_5,$$

with a orientation-reversing (the cross-cap generator) and $b_1, \dots, b_5$ peripheral; the fifth cusp word is $w_5 = a^2 b_1 b_2 b_3 b_4$. A holonomy representation is normalized with $\det \rho(a) = -1$ and $\rho(b_i) \in \mathrm{SL}(2, \mathbb{R})$ parabolic: $\mathrm{tr} \rho(b_i) = 2$ after a lift choice, and the fifth relative condition reads

$$\mathrm{tr}(A^2 B_1 B_2 B_3 B_4) = 2\sigma, \quad \sigma \in \{+1, -1\}, \quad (1)$$

where $A = \rho(a)$, $B_i = \rho(b_i)$. Since A is orientation-reversing the holonomy group is an NEC group; we use no $\mathrm{SL}(2)$ identities for words containing a beyond Cayley–Hamilton for $\det = -1$, which reads $A^2 = t_0 A + I$ with $t_0 = \mathrm{tr} A$.

Convention 2.2 (length dictionary). For a one-sided simple closed geodesic γ with holonomy A_γ scaled to $\det A_\gamma = -1$ and positive trace, Norbury’s convention [6] gives

$$\mathrm{tr} A_\gamma = 2 \sinh\left(\frac{1}{2} \ell_\gamma\right).$$

Trace functions are regular on the relative $\mathrm{GL}(2)$ character variety cut out by the parabolic conditions; identities of regular functions proved on a Zariski-dense subset hold at every character, in particular at Fuchsian ones.

Theorem 2.3 (maximal systems; Chen [1], Chen–Shen [2]). *On $N_{1,5}$ the maximum cardinality of a complete 1-system equals $1 + \binom{5}{2} = 11$; the maximum cardinality of an ordinary (pairwise at most one intersection) 1-system is also 11; every maximum ordinary system is complete and consists of one-sided curves only; and up to the full mapping class group, which may permute punctures, there is exactly one maximum system.*

A representative – the *standard watermelon* – is

$$\mathcal{C}_{\mathrm{std}} = \{\gamma_0\} \cup \{\gamma_{ij} : 1 \leq i < j \leq 5\}, \quad (2)$$

where, with the punctures in cyclic order, γ_{ij} separates the consecutive block $P_i, \dots, P_{j-1}$ in the cut disk. A van Kampen computation (elementary and re-checked at word level in the source repository) gives the oriented free-homotopy representatives

$$g_0 = a, \quad g_{ij} = a b_i b_{i+1} \cdots b_{j-1}. \quad (3)$$

All eleven curves are one-sided, so all eleven words have $\det \rho = -1$. Define the *interval-word traces*

$$t_0 = \mathrm{tr} A, \quad t_{ij} = \mathrm{tr}(A B_i \cdots B_{j-1}), \quad (4)$$

and the block traces $\beta_{ij} = \mathrm{tr}(B_i \cdots B_{j-1})$, so that $\beta_{i,i+1} = 2$.

3 The universal quad relation

Throughout this section $A \in \mathrm{GL}(2)$ has $\det A = -1$ and $B_1, B_2 \in \mathrm{SL}(2)$ are *arbitrary* – no parabolicity, no geometry. Set

$$t_0 = \mathrm{tr} A, \quad t_1 = \mathrm{tr}(A B_1), \quad t_2 = \mathrm{tr}(A B_1 B_2), \quad t_3 = \mathrm{tr}(A B_2), \quad b_1 = \mathrm{tr} B_1, \quad b_2 = \mathrm{tr} B_2, \quad k = \mathrm{tr}(B_1 B_2). \quad (5)$$

Theorem 3.1 (universal quad relation). *The seven quantities (5) always satisfy $Q = 0$, where*

$$Q := t_0^2 + t_1^2 + t_2^2 + t_3^2 + 4 + b_1 b_2 k + b_1 b_2 t_0 t_2 + k t_1 t_3 - b_1^2 - b_2^2 - k^2 - b_1(t_0 t_1 + t_2 t_3) - b_2(t_0 t_3 + t_1 t_2) - k t_0 t_2. \quad (6)$$

The polynomial Q is irreducible over $\mathbb{Q}$, has total degree 4 and sixteen terms, is quadratic and monic in each of t_0, t_1, t_2, t_3 , and is symmetric under the simultaneous swap $(t_1, b_1) \leftrightarrow (t_3, b_2)$.

Proof. Let $E = (E_0, E_1, E_2, E_3) = (I, B_1, B_2, B_1 B_2)$ and let G be the Gram matrix of the trace form, $G_{rs} = \text{tr}(E_r E_s)$. The $\text{SL}(2)$ Cayley–Hamilton identities $X^2 = \text{tr}(X)X - I$ and $\text{tr}(XY^{-1}) = \text{tr} X \text{tr} Y - \text{tr}(XY)$ express every entry in the block traces alone:

$$G = \begin{pmatrix} 2 & b_1 & b_2 & k \\ b_1 & b_1^2 - 2 & k & b_1 k - b_2 \\ b_2 & k & b_2^2 - 2 & b_2 k - b_1 \\ k & b_1 k - b_2 & b_2 k - b_1 & k^2 - 2 \end{pmatrix}. \quad (7)$$

Writing $\Delta := b_1^2 + b_2^2 + k^2 - b_1 b_2 k - 4$ for the block discriminant, direct expansion gives

$$\det G = -\Delta^2. \quad (8)$$

Let $v = (t_0, t_1, t_3, t_2)^T$, which is precisely the vector of pairings $\text{tr}(A E_r)$ (note the order: $E_2 = B_2$ pairs to t_3 and $E_3 = B_1 B_2$ pairs to t_2). Because $\det A = -1$, Cayley–Hamilton for A reads $A^2 = t_0 A + I$, whence

$$\text{tr}(A^2) = t_0^2 + 2. \quad (9)$$

Suppose first $\Delta \neq 0$. By (8) the quadruple E is then a basis of the four-dimensional space of 2×2 matrices. Expanding $A = \sum_r c_r E_r$, the coordinate vector is $c = G^{-1}v$ and $\text{tr}(A^2) = c^T G c = v^T G^{-1}v$. Combining with (9) and clearing the denominator $\det G$:

$$v^T \text{adj}(G) v - (t_0^2 + 2) \det G = 0.$$

The left-hand side factors *exactly* as

$$v^T \text{adj}(G) v - (t_0^2 + 2) \det G = -2 \Delta Q, \quad (10)$$

an identity of polynomials in the seven variables verified by direct expansion. Since $\Delta \neq 0$, this forces $Q = 0$. The locus $\Delta \neq 0$ is a nonempty Zariski-open subset of the parameter space and Q is a polynomial in the seven trace coordinates, each of which is a polynomial in the matrix entries; hence $Q = 0$ holds identically, including on $\Delta = 0$.

Irreducibility, the term count, and the monic quadratic structure are checked by exact symbolic computation (certificate `derive_quad_relation.py`, which also gives a second, basis-free proof: substituting the eight matrix entries of B_1, B_2 and the normalized $A = \begin{pmatrix} t_0 & 1 \\ 1 & 0 \end{pmatrix}$ into (6) and reducing modulo $(\det B_1 - 1, \det B_2 - 1)$ leaves remainder zero with explicit cofactors, covering $\Delta = 0$ directly). The swap symmetry is visible in (6). $\square$

Remark 3.2 (where $\det A = -1$ enters). Only through (9): for $\det A = \varepsilon$ one has $\text{tr}(A^2) = t_0^2 - 2\varepsilon$, and $\varepsilon = -1$ produces the $+2$, hence the constant $+4$ in Q . Two-sided curves satisfy an analogous identity with a different constant; the one-sided case is the one whose trace equals $2 \sinh(\ell/2)$ by [6], which is what makes Q a length relation in Section 5.

4 The Markoff–Hurwitz form

Theorem 4.1 (cusp-decorated MH form). *In the setting of Theorem 3.1, suppose additionally that the outer block is a cusp word:*

$$\mathrm{tr}(A^2 B_1 B_2) = 2\sigma, \quad \sigma \in \{+1, -1\}.$$

Then $k = 2\sigma - t_0 t_2$, and $Q = 0$ becomes

$$t_0^2 + t_1^2 + t_2^2 + t_3^2 = t_0 t_1 t_2 t_3 + b_1(t_0 t_1 + t_2 t_3) + b_2(t_0 t_3 + t_1 t_2) - 2\sigma(t_0 t_2 + t_1 t_3) - 2\sigma b_1 b_2 + b_1^2 + b_2^2. \quad (11)$$

The quartic coefficient is $c = 1$, with no rescaling of coordinates; every remaining term has total degree at most 2 in the t_i , and its coefficient is determined by the two bounding block traces b_1, b_2 and the lift sign σ .

Proof. By Cayley–Hamilton for A (Remark 3.2), $A^2 = t_0 A + I$, so $\mathrm{tr}(A^2 B_1 B_2) = t_0 \mathrm{tr}(A B_1 B_2) + \mathrm{tr}(B_1 B_2) = t_0 t_2 + k$; the cusp condition gives $k = 2\sigma - t_0 t_2$. Substitute into (6) and expand, using $\sigma^2 = 1$. The substitution creates the quartic term: $-k t_0 t_2$ contributes $+t_0^2 t_2^2$, which cancels against the $-k^2$ contribution, while $k t_1 t_3$ contributes $-t_0 t_2 t_1 t_3$; this cancellation of $t_0^2 t_2^2$ is why (11) is genuinely quartic of Markoff–Hurwitz shape rather than quadratic in two products. The bookkeeping is certified exactly in `check_mh_form.py`. $\square$

Corollary 4.2 (the Huang–Norbury quad is a special case). *For $N_{1,3}$ both blocks are single parabolics, so $b_1 = b_2 = 2$. At $\sigma = +1$ every lower-order term of (11) collapses into a perfect square and (11) becomes*

$$(t_0 - t_1 + t_2 - t_3)^2 = t_0 (-t_1) t_2 (-t_3),$$

which is the Markoff quad $(a+b+c+d)^2 = abcd$ of [4] in the signs $(a, b, c, d) = (t_0, -t_1, t_2, -t_3)$. At $\sigma = -1$ the collapse fails: the equation is $\sum_i t_i^2 - 2 \sum_{i < j} t_i t_j - t_0 t_1 t_2 t_3 - 16 = 0$, whose quadratic part is irreducible.

Proof. Substitution of $b_1 = b_2 = 2$, $\sigma = \pm 1$ in (11) and regrouping. $\square$

Thus the rank-three theory of [4] – previously an independent computation, and the frozen acceptance test of the present campaign – is a corollary of one universal identity. Note the anchor is *inhomogeneous*: degree 2 on the left, degree 4 on the right.

Theorem 4.3 (the quad system on $N_{1,5}$). *Let $\mathcal{C}_{\mathrm{std}}$ be the standard maximal system (2) with words (3) and traces (4). For every triple $i < j < l$, the three curves $\{\gamma_{ij}, \gamma_{il}, \gamma_{jl}\}$ are pairwise once-intersecting and one-sided, and the relation $Q = 0$ of (6) holds with*

$$(t_1, t_2, t_3) = (t_{ij}, t_{il}, t_{jl}), \quad (b_1, b_2, k) = (\beta_{ij}, \beta_{jl}, \beta_{il}).$$

Moreover $\beta_{i,i+1} = 2$ for $i = 1, \dots, 4$, and by Cayley–Hamilton together with the fifth cusp condition (1),

$$\beta_{15} = 2\sigma - t_0 t_{15}. \quad (12)$$

Exactly three triples – $(1, 2, 5)$, $(1, 3, 5)$, $(1, 4, 5)$ – have β_{15} in the k slot and therefore satisfy the genuine Markoff–Hurwitz equation (11) with quartic coefficient 1 and bounding blocks

$$(b_1, b_2) = (\beta_{1j}, \beta_{j5}), \quad j \in \{2, 3, 4\}.$$

The other seven triples have $k = \beta_{il}$ with $(i, l) \neq (1, 5)$, not cusp-determined; their relations are cubic in the traces and carry no quartic product term.

Proof. Apply Theorem 3.1 verbatim with $B_1 = B_i \cdots B_{j-1}$ and $B_2 = B_j \cdots B_{l-1}$: its proof used no consecutivity, parabolicity, or geometricity of the blocks. For the triple $(1, j, 5)$ the outer block is $B_1 \cdots B_4$, and $\text{tr}(A^2 B_1 B_2 B_3 B_4) = \text{tr} \rho(w_5) = 2\sigma$ is precisely (1); Theorem 4.1 applies. Identity (12) is $\text{tr}(A^2 B_1 \cdots B_4) = t_0 \text{tr}(A B_1 \cdots B_4) + \text{tr}(B_1 \cdots B_4) = t_0 t_{15} + \beta_{15}$. Adjacent blocks are single parabolics, giving $\beta_{i,i+1} = 2$. All ten instantiations, and the adjacent-block and Cayley–Hamilton bookkeeping, are additionally certified by exact symbolic expansion on a normalized chart in `check_quad_system.py`. $\square$

Remark 4.4 (the decorations are intrinsic). The auxiliary block traces $\beta_{13}, \beta_{14}, \beta_{24}, \beta_{25}, \beta_{35}$ appearing as coefficients are themselves determined by the eleven curve traces: they are explicit rational functions of (t_0, t_{ij}) on the trace image (proved in the source ledger via the pentagon elimination; see Section 7). The system of ten quads is thus a relation system among the eleven lengths and nothing else. At an exact rational point off the genericity divisor its Jacobian in the sixteen coordinates $(t_0, t_{ij}, \beta.)$ has rank 9, so the quads cut out a variety of local dimension seven – equal to the trace-image dimension – and an exact Gröbner elimination refutes the false local branch left open by earlier resultant methods (COMPUTED; certificates `check_quad_system.py`, `check_quad_rigidity.py`).

5 From traces to lengths

Theorem 5.1 (MH as an exact length identity). *Let $\{\gamma_0, \gamma_{1j}, \gamma_{15}, \gamma_{j5}\}$, $j \in \{2, 3, 4\}$, be a cusp-decorated one-sided triple of $\mathcal{C}_{\text{std}}$ together with γ_0 , with geodesic lengths $\ell_0, \ell_1, \ell_2, \ell_3$ in a complete finite-area hyperbolic metric on $N_{1,5}$. Put $u_m = 2 \sinh(\ell_m/2)$. Then there are signs $\delta_m \in \{\pm 1\}$ such that $x_m = \delta_m u_m$ satisfies (11) exactly, with b_1, b_2 the traces of the bounding block words and σ the fifth-cusp lift sign.*

Proof. By [6], a one-sided closed geodesic has $\text{tr} A_\gamma = 2 \sinh(\ell_\gamma/2)$ once A_γ is scaled to determinant -1 and positive trace; hence each trace equals $\delta_m u_m$ for a sign fixed by the lift and orientation choices. Theorem 4.1 via Theorem 4.3 is an identity of regular functions on the relative character variety, so it holds at the character of the Fuchsian holonomy. (No analytic continuation is needed for this direction; for context on the Fuchsian locus inside the character variety see Huang [3].) $\square$

The signs δ_m and the lift σ are artifacts of coordinates. They are removed by a *sign norm*.

Lemma 5.2 (sign norm). *For a nonzero polynomial $L \in R[z_1, \dots, z_k]$ over an integral domain R , define*

$$\mathcal{N}(L)(z_1, \dots, z_k) = \prod_{\epsilon \in \{\pm 1\}^k} L(\epsilon_1 z_1, \dots, \epsilon_k z_k).$$

Then $\mathcal{N}(L)$ is a nonzero polynomial, invariant under $z_i \mapsto -z_i$ for each i , and there is a unique nonzero $\widehat{L} \in R[s_1, \dots, s_k]$ with $\mathcal{N}(L)(z) = \widehat{L}(z_1^2, \dots, z_k^2)$. If a tuple $(x_1, \dots, x_k)$ satisfies $L(x) = 0$, then $\widehat{L}(x_1^2, \dots, x_k^2) = 0$.

Proof. Each flip $z_i \mapsto -z_i$ permutes the 2^k factors, so $\mathcal{N}(L)$ is invariant; a polynomial invariant under every single-variable sign flip is even in each variable separately, hence a polynomial in the squares, uniquely so. A product of finitely many nonzero polynomials over an integral domain is nonzero. The last claim is immediate since one factor of $\mathcal{N}(L)$ vanishes at x . $\square$

Theorem 5.3 (uniform sign-free length relations). *Fix a cusp-decorated triple as in Theorem 5.1 and let $L_\sigma(t_0, t_1, t_2, t_3)$ denote the Markoff–Hurwitz polynomial of (11) with its block-trace coefficients. Then, writing $s_m = 4 \sinh^2(\ell_m/2)$, every complete finite-area hyperbolic metric on $N_{1,5}$ satisfies*

$$\widehat{L}_\sigma(s_0, s_1, s_2, s_3) = 0,$$

a nonzero polynomial identity in the s_m with coefficients polynomial in the block traces and independent of all orientation and sign choices of the four curves. The same construction applied to the seven undecorated quads of Theorem 4.3 yields sign-free relations for every one-sided triple. Moreover every maximal complete 1-system on $N_{1,5}$ admits an ordering for which these identities hold verbatim.

Proof. By Theorem 5.1 the tuple $x_m = \delta_m u_m$ satisfies $L_\sigma(x) = 0$; Lemma 5.2 applied in the four variables t_m (block traces regarded as coefficients) gives $\widehat{L}_\sigma(u_0^2, \dots, u_3^2) = 0$, and $u_m^2 = s_m$. For the last claim, Theorem 2.3 supplies a mapping class carrying any maximal system to $\mathcal{C}_{\text{std}}$; pulling the marked metric through it moves the point of Teichmüller space, over all of which the identities are uniform, and a puncture permutation only relabels peripheral generators. $\square$

Remark 5.4. The identity $\widehat{L}_\sigma = 0$ is the honest, coordinate-free answer to the question “what equation do the lengths satisfy”: the quantities $s_m = 4 \sinh^2(\ell_m/2)$ are intrinsic to the unoriented geodesics, no lift of the holonomy is chosen, and the two lift signs $\sigma = \pm 1$ can be handled symmetrically by taking the product over both, at the cost of degree. The source repository also certifies four compact sign-free relations among all eleven traces obtained by the same mechanism from resultant identities; they are not needed for the results above.

6 Vieta involutions and the rank-five dichotomy

On $N_{1,3}$, the single Markoff quad carries Vieta *flips* which generate the mapping class dynamics on characters [4], exactly as in [5]. Half of this picture survives at rank five, and half provably fails.

Proposition 6.1 (Vieta structure; the surviving half). *Q is quadratic and monic in each one-sided trace, so fixing the other six coordinates and taking the second root defines four involutions*

$$\begin{aligned} V_0 : t_0 &\mapsto b_1 t_1 + b_2 t_3 + (k - b_1 b_2) t_2 - t_0, & V_1 : t_1 &\mapsto b_1 t_0 + b_2 t_2 - k t_3 - t_1, \\ V_2 : t_2 &\mapsto b_1 t_3 + b_2 t_1 + (k - b_1 b_2) t_0 - t_2, & V_3 : t_3 &\mapsto b_1 t_2 + b_2 t_0 - k t_1 - t_3, \end{aligned}$$

each fixing (b_1, b_2, k) . Each V_i maps Q to Q identically – not merely up to a factor, and not merely on the zero locus – hence is a polynomial automorphism of affine space preserving the quad hypersurface; this is the Markoff–Hurwitz automorphism signature of [5]. Two of the four are realized by matrix words:

$$V_1 : t'_1 = \text{tr}(AB_2^{-1}B_1B_2), \quad V_3 : t'_3 = \text{tr}(AB_1B_2B_1^{-1}),$$

verified by exact reduction modulo $(\det B_1 - 1, \det B_2 - 1)$. All six pairwise products $V_i V_j$ have order greater than 12, consistent with the infinite Markoff–Hurwitz automorphism group.

Proof. That V_i is an involution preserving Q identically is a polynomial identity, verified exactly (certificate `check_vieta.py`); monicity in each t_i is visible in (6), and the displayed

linear coefficients are read off the mixed terms. For the word realizations, expand both sides by the $\text{SL}(2)$ trace calculus and reduce exactly. (The substitution $B_1 \mapsto B_2^{-1}B_1B_2$ fixes b_1, b_2, k, t_3 but not t_2 ; the displayed word realizes only the t_1 -component of V_1 , so no identification of V_1 with a mapping class is claimed – the next result shows this caution is essential.) $\square$

Proposition 6.2 (the failing half: no naive global action). *The involutions of Proposition 6.1, applied triple-wise, do not assemble into an action on the eleven-tuple of traces. Concretely, at a frozen exact rational point of the trace image, applying V_1 inside the triple $(1, 2, 3)$ – the moved coordinate being $t_{12} \mapsto \beta_{12}t_0 + \beta_{23}t_{13} - \beta_{13}t_{23} - t_{12}$, numerically $7 \mapsto \frac{119}{2}$ – while holding the other ten traces fixed:*

1. *leaves nonzero quad residuals in the triples $(1, 2, 4)$ and $(1, 2, 5)$ when the auxiliary block traces are held fixed; and*
2. *admits no repair: allowing all five unknown block traces to vary, the exact lexicographic eliminating Gröbner basis of the ten quad equations at the moved tuple is $\{1\}$ – the empty variety – whereas at the original point the same elimination is solvable and pins every block trace.*

Since the quad system is necessary for membership (Theorem 4.3), the moved tuple lies outside the trace image. (COMPUTED; exact certificate `check_vieta_incompatibility.py`, data `vieta_incompatibility.json`.)

Remark 6.3 (interpretation). Whatever mapping-class dynamics the maximal system carries, it is *not* the naive triple-wise Vieta action – in contrast with $N_{1,3}$, where a single quad carries the whole structure and the flips are complete. The ten quads form a coupled *pentagon*: five monic quadratics linked by five bilinear couplings in a 5-cycle (established in the source ledger), and the coupling does not transport a Vieta conjugate of one quad to a root of its neighbour, so a flip dies at the first edge. This is the precise sense in which Markoff–Hurwitz structure at rank five is a decorated *family* rather than one equation: the positive results of Sections 4–5 and this obstruction are two halves of one closed answer. Whether some corrected simultaneous move – necessarily adjusting several coordinates at once – realizes mapping-class dynamics on the eleven-tuple remains an open question, as does the determination of the full ideal of relations among the eleven traces; neither affects the statements proved here.

7 Provenance, verification model, and dependency disclosure

Provenance

The mathematics reported here was produced by an autonomous multi-agent AI research fleet between 2026-08-04 and 2026-08-18 (fleet window 02-markoff-hurwitz), working from a task specification whose frozen acceptance test was the recovery of the Huang–Norbury quad on $N_{1,3}$ – now Corollary 4.2. The campaign repository contains a tiered claim ledger, an adversarial referee record, a negative-results file (including the documented mid-campaign correction that the holonomy is an NEC group, after which every convention was re-derived), and frozen exact-computation certificates. The equality-of-polynomials steps cited as certificates use exact integer and rational arithmetic only: `derive_quad_relation.py` (Theorem 3.1: identity (10), irreducibility, and the basis-free second proof), `check_mh_form.py` (Theorem 4.1), `check_quad_system.py` (Theorem 4.3;

Remark 4.4), `check_vieta.py` (Proposition 6.1), and `check_vieta_incompatibility.py` (Proposition 6.2). Statements labelled COMPUTED are finite exact computations at explicitly recorded rational points and are reproducible from the frozen code and data; every other proof above is conventional and self-contained.

Load-bearing literature inputs

Used as cited, without in-house reproof: Chen [1] and Chen–Shen [2] for Theorem 2.3 (maximality 11, one-sidedness, completeness, and uniqueness under the full mapping class group – used only in Theorem 4.3 through the standard system and in the final clause of Theorem 5.3); Norbury [6] for the one-sided length dictionary $\text{tr} = 2 \sinh(\ell/2)$ (Theorem 5.1); Huang–Norbury [4] for the $N_{1,3}$ anchor recovered in Corollary 4.2 (used as a cross-check, not as an input); Hu–Tan–Zhang [5] for the Markoff–Hurwitz automorphism context of Proposition 6.1 (context, not input); and Huang [3] for the placement of the Fuchsian locus in the character variety (contextual in Theorem 5.1; the direction used there needs only that trace identities are polynomial identities on the character variety). Theorems 3.1 and 4.1 and Lemma 5.2 use no literature input at all.

Scope

This paper deliberately contains only the closed chain. The same campaign also established, with exact certificates: that no polynomial relation of total degree ≤ 4 holds among the eleven traces; that 34 specific symmetric Markoff–Hurwitz-type supports in all eleven lengths – including the literal analogue $e_1^2 = c e_{11}$ of the $N_{1,3}$ anchor – admit no uniform constant; that the ten quads do *not* generate the prime ideal of all relations (an explicit degree-ten element lies outside), with $6 \leq \mu(I) \leq 10$ for its minimal number of generators; and that the coordinate S_5 -action is not a symmetry of the trace image while the order reversal $i \mapsto 6-i$ is. Determining the full relation ideal, and deciding whether any Markoff–Hurwitz-type equation in all eleven lengths exists, remain open problems and are claimed nowhere in this paper.

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